

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 2

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO2

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 2

Unit Title: Partition-Based Clustering

Unit Mapped RBT: L4 (Analyze)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Analyze the assumptions that k-Means makes about the shape and the size of the groups it looks for. (10 Marks)

- (a) Analyze the shape of the region each cluster occupies, and explain why it takes that shape.
- (b) Infer two kinds of dataset on which the method will fail, giving the reason in each case.

[L4 - Analyze]

Q2. Differentiate hard clustering from soft clustering. (10 Marks)

- (a) Differentiate what each of the two assigns to a single record.
- (b) Infer a situation in which a soft assignment carries information that a hard one throws away.

[L4 - Analyze]

Q3. Distinguish inertia, the silhouette coefficient and the Davies-Bouldin Index as measures of a clustering. (10 Marks)

- (a) Distinguish what each of the three actually measures about a partition.
- (b) Infer why inertia alone cannot be used to choose the number of clusters while the other two can.

[L4 - Analyze]

Q4. Analyze the influence of the starting centres on the partition that k-Means finally reports. (10 Marks)

- (a) Analyze why a poor set of starting centres leads to a poor final partition.
- (b) Infer how k-Means++ and multiple restarts each reduce that risk, and what neither of them can guarantee.

[L4 - Analyze]

Q5. Infer what a very low inertia combined with a very low silhouette score tells you about a clustering. (10 Marks)

- (a) Analyze what each of the two readings measures about the partition.
- (b) Infer the most likely explanation for the combination, and state what you would examine next.

[L4 - Analyze]

Q6. Differentiate MiniBatch k-Means from the full k-Means algorithm. (10 Marks)

- (a) Differentiate the two by what each one uses to update the cluster centres.
- (b) Infer what is gained and what is given up, and state when the exchange is worth making.

[L4 - Analyze]

Q7. Analyze why standardising the features changes the clusters that k-Means reports. (10 Marks)

- (a) Analyze the effect that one wide-ranging unscaled column has on the distances.
- (b) Infer how the cluster profiles would differ before and after the columns are standardised.

[L4 - Analyze]

Q8. Distinguish the objective that k-Means minimises from the objective that k-Medoids minimises. (10 Marks)

- (a) Distinguish the two by the quantity each one sums over the records of a cluster.
- (b) Infer which of the two is the more disturbed by a small number of extreme records, and give the reason.

[L4 - Analyze]

Q9. Analyze the elbow method as a way of deciding the number of clusters. (10 Marks)

- (a) Analyze why the inertia curve always falls, and state what the bend in it actually represents.
- (b) Infer two reasons the method can fail to give a clear answer on real data.

[L4 - Analyze]

Q10. Relate the features supplied to a k-Means clustering to the meaning of the segments it reports. (10 Marks)

- (a) Analyze what the algorithm is and is not able to see about the customers.
- (b) Infer what a manager should be told when the segments produced cannot be described in business terms.

[L4 - Analyze]

----- End of Sample Question Set -----